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Numerical analysis of hemodynamic changes in the left atrium due to atrial fibrillation

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ABSTRACT

Atrial fibrillation (AF) disrupts movement of the left atrium (LA) and worsens the vital prognosis by causing thromboembolism. Ultrasound Doppler measurement, phase-contrast magnetic resonance imaging (PC MRI), as well as computational fluid dynamics (CFD) have revealed hemodynamic changes in the LA due to AF, such as stagnation of blood flow in the left atrial appendage (LAA). However, quantitative evaluation of the hemodynamics during AF has not been conducted, and the effects of important AF characteristics, such as a lack of active contraction of the LA (atrial kick) in late diastole and the occurrence of high-frequency fibrillation (> 400 bpm) of the atrial wall, on blood flow field and concomitant hemodynamic stresses have not been completely understood. In this study, the effects of the above-mentioned two characteristic phenomena of AF on blood flow and hemodynamic parameters were quantitatively investigated. Based on MRI of a healthy volunteer heart, one healthy LA model and two AF models (one without atrial kick, and one without atrial kick and with high-frequency fibrillation) were constructed to perform hemodynamic analysis, and the computational results were compared. The results revealed that each characteristic phenomenon of AF influenced hemodynamics. Especially, atrial wall movement by high-frequency fibrillation had a large impact on the stagnation of blood flow. The relative residence time (RRT), which is an indicator of stagnation of blood flow, increased in the upper part of the LAA during AF. This result implies that there is a local thrombus-prone site in LAA when AF occurs.
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1. Introduction

The left atrium (LA) is located at the upper dorsal part of the heart and has a characteristic part called the left atrial appendage (LAA), which bulges in the inner forward direction (Al-Saady et al., 1999). It pumps arterial blood from the four bilateral pulmonary veins (PVs) and ejects the blood toward the left ventricle (LV) through the mitral valve (MV) by both passive and active motions synchronized with ventricular expansion and contraction (Pagel et al., 2003). Atrial fibrillation (AF) is the most common arrhythmia which disrupts the movement of the LA (Iwasaki et al., 2011). The important characteristics of AF can be summarized as a lack of an active contraction of the LA (atrial kick) in late diastole and the occurrence of high-frequency fibrillation (> 400 bpm) of the atrial wall. AF may cause no symptoms but is often associated with heart palpitation. Except for patients with severe underlying disease or hypofunction of LV contraction, aggressive treatments are not performed. However, it has been reported that thromboembolism derived from AF worsened the vital prognosis (Benjamin et al., 1998). An increase in the incidence of AF due to aging has also been suggested to be a cause of severe cerebral infarction (Wolf et al., 1991; Yamanouchi et al., 1989). Therefore, diagnosis, treatment, and management of AF are clinically important issues.

AF is diagnosed mainly by electrocardiogram (ECG). In order to diagnose the causes and complications of AF, morphological approach for entire heart, including LA and LAA, is conducted by means of computed tomography or ultrasound imaging, in addition to a blood test. On the other hand, there have been trials to utilize hemodynamic information on the LA obtained by phase-contrast magnetic resonance imaging (PC MRI) (Fluckiger et al., 2013; Fyrenius et al., 2001; Kilner et al., 2000) and ultrasonic measurement (Nakai et al., 2007; Nishimura et al., 1990; Park et al., 2013) for development of therapeutic strategy. Computational fluid dynamics (CFD) (Zhang and Gay, 2008) and particle image velocimetry (PIV)

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(Mouret et al., 2004) have also revealed morphological changes of vortices in the LV due to AF and stagnation of blood flow in the LAA leading to formation of thrombosis. However, though hemodynamics in LA during sinus rhythm and AF were qualitatively compared, no quantitative evaluations were conducted. Moreover, effects of each phenomenon caused by AF, i.e., the lack of an atrial kick and the occurrence of high-frequency fibrillation, on the blood flow field and hemodynamic stresses have not been completely understood.

In this study, effects of the above-mentioned two characteristics of AF on blood flow and hemodynamic parameters were quantitatively investigated. Hemodynamic analyses were performed using healthy and diseased LA models, and their computational results were compared.

2. Methods

2.1. Magnetic resonance imaging

This study was approved by the local ethics committees, and informed consent was obtained from the participating volunteer. First, for measurement of variation of the LA morphology, the heart of the 23-year-old healthy male volunteer was measured using a 1.5 T MRI scanner system (Achieva 1.5T, Philips Electronics, Holland) with a SENSE cardiac coil and a two-dimensional balanced turbo field echo at Miyagi Cardiovascular and Respiratory Center. Four chamber T1 weighted images were acquired on 12 slices at intervals of 5 mm with the following parameters: repetition time (TR), 3.3 ms; echo time (TE), 1.7 ms; flip angle (FA), 60°; field of view (FOV), 400 mm; and matrix size, 256 × 256 pixel. The image sets were obtained at 20 phases during the R–R interval (one cardiac cycle). Heart rate of the subject was 91 bpm (T=659 ms), so that 15 s was needed to obtain sequential images on one cross section. The total scan time was 6 min. Next, measurement of time-varying blood flow velocity in the LA was performed using a 1.5 T MRI scanner system (Excelart 1.5T, Toshiba Medical Systems, Japan) with a torso SPEEDER coil and phase shift magnetic resonance (MR) angiography at Tohoku Kosei Nenkin Hospital. All three velocity components were measured in the supine position in a four-chamber view. Thirteen slices were obtained at intervals of 5 mm, dividing one cardiac cycle into eight phases. The other parameters were set as follows: TR=24 ms, TE=10 ms, FA=20°, FOV=350 mm, matrix size=256 × 256 pixel, and VENC=1.0 m/s. The PC MRI data were obtained within 100 min. In each imaging, thoracic movement was reduced by breath-holding to suppress artifacts.

2.2. Left atrial model

As a basal model of computational models for numerical analysis, a three-dimensional (3D) morphology of the LA was reconstructed from a set of MR slice images in early diastole by using commercial software (Mimics 7.3, Materialise, Belgium). Fig. 1(a) shows the reconstructed left heart in stereolithography (STL) format: the red part is the LA, and the gray part is the LV. The LA has four PV inlets (PV1–PV4) and one MV outlet, and the LAA exists near PV3. Since the shape of the MV was not clearly identified in the MR images, it was replaced with a circle 24 mm in diameter (Ormiston et al., 1981). Cross-sectional areas of PVs and the MV in the computational model were as follows: PV1, 36.7 mm^2; PV2, 37.1 mm^2; PV3, 12.6 mm^2; PV4, 33.7 mm^2; and MV, 451.0 mm^2.

Deformation of the healthy LA model (HA model) in one cardiac cycle was obtained based on the location of the MV annulus in the MR slice images. As shown in Fig. 1(b), the xy coordinate system was defined with the origin at the upper-left position of the MR image. In addition, another $x'y'$ coordinate system was defined by setting the y' axis in the moving direction of the MV, and the origin was set at point (x_{org}, y_{org}) of little displacement. The positions of the left-side annulus of the MV were manually extracted on four of 12 MR slice images at each measured phase. The corresponding sequential positions at two different phases (see (x_{m1}, y_{m1}) and (x_{m2}, y_{m2}) in Fig. 1(b)) were then picked up and converted into the $x'y'$ coordinate system. For simplification, the right-side annulus of the MV was symmetrically determined by folding the one on the left side over the y'-axis. The movement of the lower part of the LA wall ($y' \geq 0$) in the y'-direction was computed with the assumption of a simple compression/extension, whereas that in the x'-direction was computed by linearly increasing the compression rate toward the MV. In contrast, the upper part of the LA ($y' < 0$) was fixed. Finally, a new morphology of the LA was obtained by inversely transforming the moved atrial wall in the $x'y'$ coordinate system into the xy coordinate system. Here, to continuously and smoothly change the LA morphology, a linear interpolation was applied between the slices. Moreover, an interpolation in the time domain was applied with a third-order spline interpolation based on the position of MV measured by MRI at 20 phases to divide one cardiac cycle into 200 computational time steps.

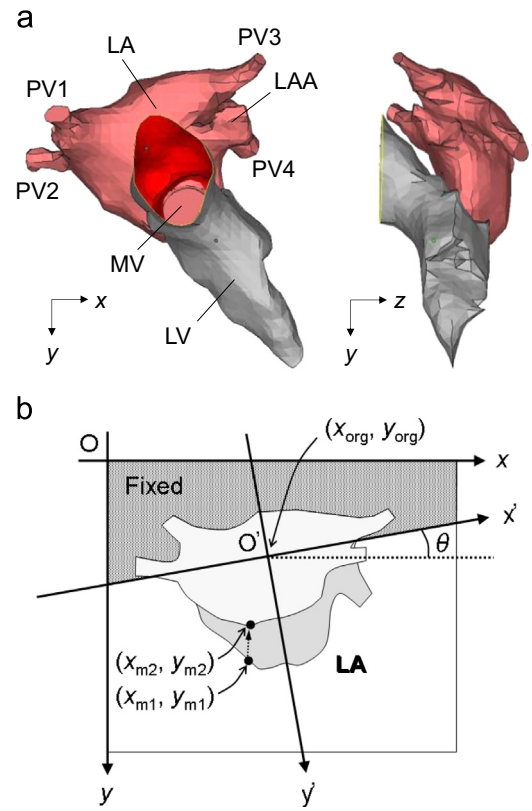

Fig. 1. A healthy left heart reconstructed from MRI: (a) 3D morphologies of the left atrium (LA, the red part) and the left ventricle (LV, the gray part), and (b) the definition of the coordinate systems. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The MR slice image and the 3D morphology of the computational model are compared in Fig. 2 at four different time steps at t=0.033, 0.165, 0.330, and 0.494 s, corresponding to the computational time step of 10, 50, 100, and 150, respectively. The yellow outline superimposed on each MR image displays the corresponding cross section of the computational model. The blue and red cross sections in the computational models show PVs and the MV, respectively. The cross-sectional shapes of the HA model almost completely agree with those in the MR image though a gap is observed beneath the left PV at peak systole (t=0.330 s, see a white arrow). This implies the validity of the deformation estimated by the location of the MV annulus. Fig. 3 represents the variation of LA volume in the HA model, in which the MV closes in $t > 0.33$ s and an atrial kick showed a volume decrease in $0.25\ \text{s} \leq t \leq 0.33$ s. The time-varying volume of the LA was reproduced with the sufficiently small interval of time step with temporal interpolation of the MRI data.

Computational models mimicking AF were constructed based on the HA model since it was difficult to obtain MR slice images of an actual AF patient due to the limitation of time resolution of MRI. In this study, a model without an atrial kick (AF_0 model) and another model without an atrial kick and with high-frequency fibrillation (AF_1 model) were created as shown in Fig. 3. It is noted that the AF_0 model is different from the actual LA with AF, in which the lack of the atrial kick and high-frequency fibrillation occur simultaneously. However, the model was used as a comparative model to investigate the effects of each phenomenon of AF on hemodynamics. High-frequency fibrillation depends on the patient and the severity. In the AF_1 model, high-frequency fibrillation was modeled by a sine wave of 7.58 Hz (five times the heart beat) with an amplitude of 2.35% of the difference between the maximum and minimum volumes of the HA model.

2.3. Computational methods

The same algorithms and conditions were applied to computational analyses with the HA model and the AF_0 and AF_1 atrial fibrillation models. Density ρ and viscosity μ of blood were set within the normal range as 1050 kg/m^3 and 3.5×10^{-3} Pa s, respectively. Regarding boundary conditions, two conditions were switched between diastole and systole. In diastole ($0\ \text{s} \leq t \leq 0.33$ s), values of pressure at four inlets of PVs and one outlet of the MV were assumed to be constant at 10 mmHg and 8 mmHg, respectively (Smith and Kampine, 1980). Because of the pressure difference and the wall motion of the LA shown in Fig. 3, blood flowed through the LA. In systole ($0.33\ \text{s} < t \leq 0.66$ s), solid wall condition was assumed at the MV, which was close, while a constant pressure of 10 mmHg was maintained at

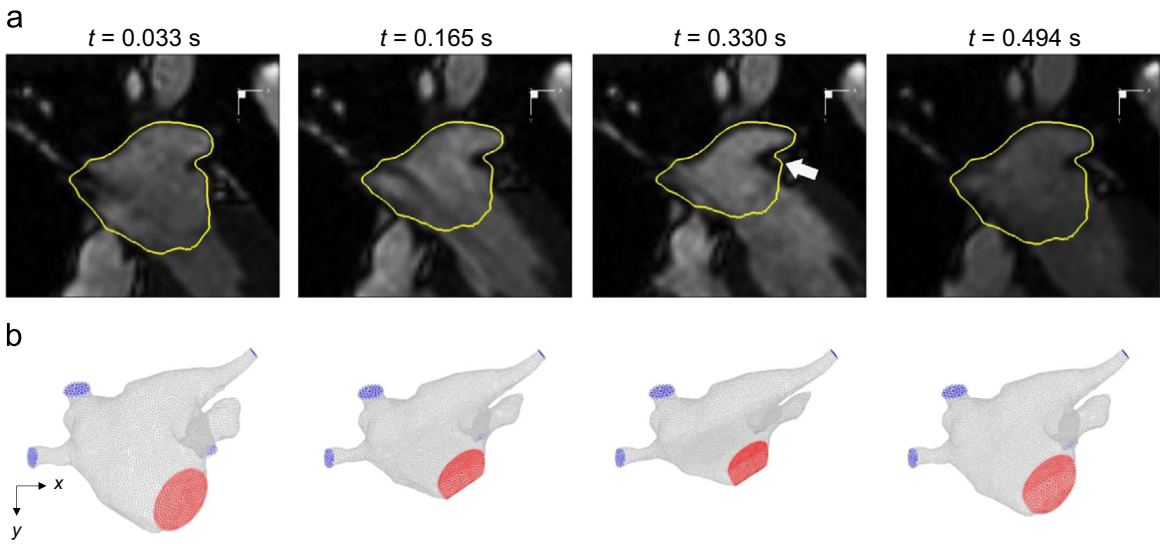

Fig. 2. Comparison of variations of the healthy LA between (a) MR slice images and (b) the computational model. Each yellow outline represents the corresponding cross section of the computational model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

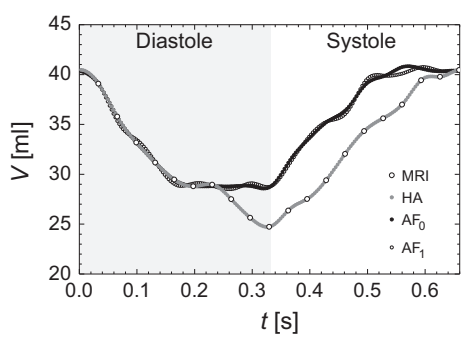

Fig. 3. Volume change of the LA: MRI data and the three computational models (HA, AF_0, and AF_1).

the four inlets of PVs. Consequently, the blood was pumped into the LA due to the wall motion of the LA. The MV was considered to instantly open and close, ignoring movements of the MV leaflets. The initial condition was the computational result of a steady flow in LA at $t=32.95$ ms with the above-mentioned boundary conditions, and each unsteady hemodynamic analysis was started at that time.

The STL model of LA was imported into commercial computational grid generation software (Gambit 2.2, ANSYS, USA), and tetrahedral computational elements were generated. The total number of computational elements was set at 149,330 after preliminary computations, a compromise between computational accuracy and computational time. Commercial CFD software (FLUENT 6.3, ANSYS, USA) was employed, and unsteady hemodynamic analyses were performed by dividing one cardiac cycle into 200 time steps ($\Delta t=3.295$ ms). The residual for convergence was set at 10×10^{-4}. Each computation was carried out in four cardiac cycles after a transient of nearly one cycle in order to obtain a quasi-periodic solution, and the solution in the last cycle was used for further evaluation (Supplementary Fig. S1). Three hemodynamic parameters derived from wall shear stress (WSS) acting on the LA wall, i.e., the time-averaged wall shear stress (TAWSS), oscillatory shear index (OSI) (Ku et al., 1985), and relative residence time (RRT) (Himburg et al., 2004) were compared between the computational models.

3. Results

Variations of blood flow volume at the MV and PV4 are compared in Fig. 4. In Fig. 4(a), the waveform at the MV in the HA model has two peaks: one is the E wave due to relaxation of LV in the early diastole, and the other one is the A wave due to the atrial kick in late diastole. On the other hand, the A wave is not observed in the AF_0 and AF_1 models, and a fluctuation by high-frequency fibrillation is observed in the AF_1 model. The E/A peak velocity ratio and the mean blood flow volume at the MV are shown

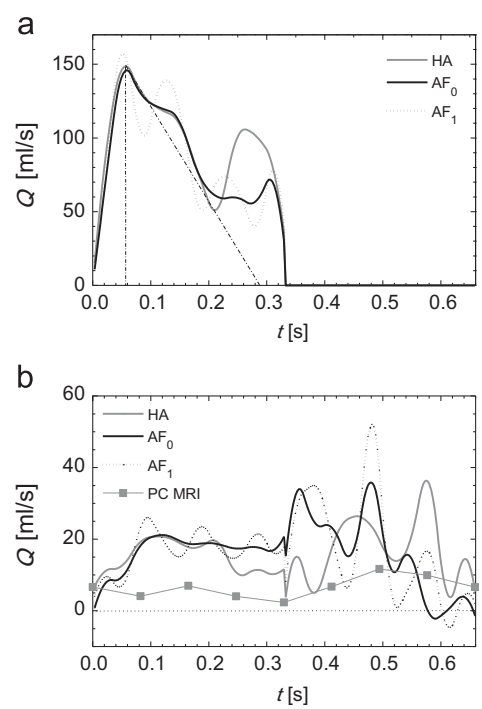

Fig. 4. Variations of blood flow volume at (a) MV and (b) PV4.

Table 1
Parameters related to hemodynamics in each LA model: the E/A peak velocity ratio and the mean blood flow volume at the MV; the hemodynamic parameters on the LAA, the space-averaged values of the TAWSS, OSI, and RRT, the minimum value of the TAWSS, and the maximum values of the OSI and the RRT.

Computational model	HA	AF_0	AF_1
E/A peak velocity ratio	1.40	2.47	2.13
Mean blood flow volume [ml]	32.04	28.68	28.78
TAWSS [Pa]	0.31	0.32	0.35
OSI [–]	0.12	0.12	0.20
RRT [Pa^{-1}]	6.38	5.98	7.94
$TAWSS_{min}$ [Pa]	0.05	0.05	0.07
OSI_{max} [–]	0.43	0.42	0.47
RRT_{max} [Pa^{-1}]	51.05	36.05	141.78

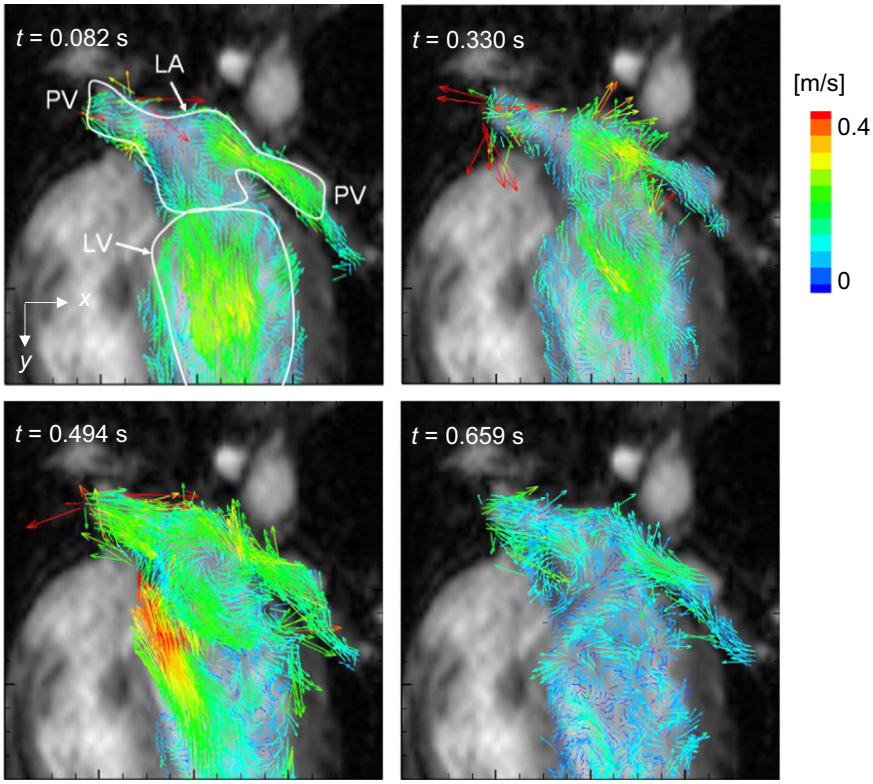

Fig. 5. Velocity vectors of blood flow on a cross section of the healthy LA measured by PC MRI.

in Table 1. Because of the lack of the atrial kick in the AF_0 and AF_1 models, the E/A ratio increases, and the total blood flow volume at the MV in one cardiac cycle decreases by 10.5% and 10.2% of that in the HA model, respectively. In Fig. 4(b), the waveforms of the blood flow volume at PV4 are qualitatively similar between the HA model and PC MRI though quantitative differences exist. There are few differences between the healthy and diseased models in early diastole, but changes due to AF are observed after the atrial kick ($t \geq 0.25$ s).

Velocity vectors obtained by PC MRI on a cross section of the left heart at four times are shown in Fig. 5. In diastole ($t=0.082$ s), blood in the LA simply flows from bilateral PVs to the MV. At the terminal phase of diastole ($t=0.330$ s) just before the MV closes, blood flow slows down, showing a decreased inflow from the PVs. Then, in systole ($t=0.494$ s), blood rapidly flows into the LA and forms a counterclockwise swirl. In the same image, ejection of blood from the LV to the aorta is displayed. Finally, at the terminal phase of systole ($t=0.659$ s), blood flow velocity decreases, and the center of the swirl moves to the left-hand side in the image (the right PV side).

At diastole and systole, streamlines departing from the four PVs (PV1–PV4) in each computational model are depicted in Fig. 6. Note that the results in systole (the right column) are shown when the volume of LA is approximately the same ($V \approx 35.5$ ml). In the HA model, streamlines show qualitatively the same feature as the above-mentioned velocity vectors measured by PC MRI in Fig. 5. No remarkable change of streamlines was recognized between the computational models in diastole, but changes of the vortices were observed in systole. Moreover, in the AF_0 and AF_1 models, there were phases with disturbed blood flow, small vortices, and the disappearance of streamlines, indicating stagnation of blood flow or regurgitation.

Fig. 7 shows WSS distributions on the computational models at the same times in diastole and systole as in Fig. 6. The time-dependent changes of WSS distribution are similar to each other

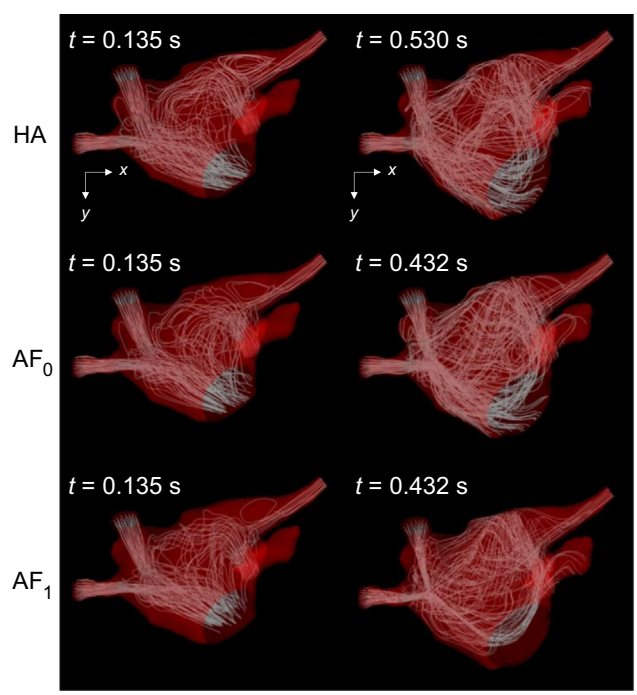

Fig. 6. Streamlines in the healthy model (HA) and the diseased models (AF_0 and AF_1) at diastole (the left column) and systole (the right column). The results in systole are shown when the volume of LA is approximately same ($V \approx 35.5$ ml).

on the whole, indicating small WSS on the LAA during one cardiac cycle. However, in contrast to the HA model, the diseased models showed an increase of WSS at the upper wall, where blood flow impinges, at an earlier time owing to the lack of the atrial kick.

Distributions of hemodynamic parameters on the LA are compared in Fig. 8. There is no significant difference between distributions of

TAWSS by three computational models (the left column). It is notable that TAWSS on the LAA is small in all models. The OSI value becomes large at the inlet of the LAA and on the upper wall of LA in the AF_0 model in comparison with that in the HA model (the middle column). Moreover, the OSI increases on the lower wall of the LA and the LAA in the AF_1 model. Although the value of RRT shows no significant difference between the HA and AF_0 models, it becomes specifically large at the upper site of the LAA in the AF_1 model (the right column).

For more detailed observation of the hemodynamic parameters on the LAA, enlargements of distributions of hemodynamic parameters on the LAA viewed from two different points are displayed in Fig. 9. In the images, the positions of the minimum TAWSS ($TAWSS_{min}$), the maximum OSI (OSI_{max}), or the maximum RRT (RRT_{max}) are indicated with white arrows, and those values are summarized in Table 1 with the mean values of hemodynamic parameters on the LAA. The TAWSS distributions are almost identical between the three computational models, but the distributions of the OSI and RRT are different. Particularly, the AF_1 model indicates OSI_{max} at a position different from those of the other models. Also, RRT_{max} is 2.8 times larger than that in the HA model (Table 1) at a different position. In the AF_1 model, both mean values of the OSI and the RRT increase more than in the other models.

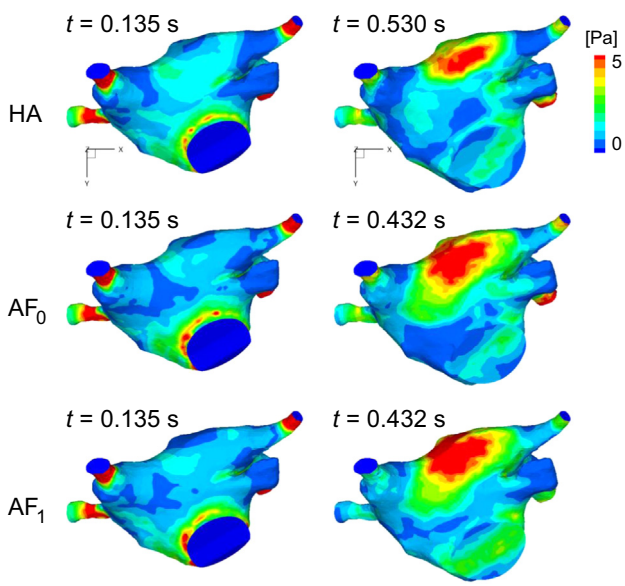

Fig. 7. WSS distributions on the healthy model (HA) and the diseased models (AF_0 and AF_1) at diastole (the left column) and systole (the right column). The results in systole are shown when the volume of LA is approximately same ($V \approx 35.5$ ml).

4. Discussion

The results demonstrated that each characteristic phenomenon of AF, i.e., the lack of an atrial kick and the occurrence of high-frequency fibrillation, influenced the blood flow field and hemodynamic stresses. Especially, AF caused a local stagnation of blood flow in the LAA. As clearly indicated by streamlines in Fig. 6, the LAA is a thrombosis-prone site because it is located at a distal site of the mainstream of blood from PVs to the MV, resulting in that blood entering into the LAA is difficult (Al-Saady et al., 1999). Blood flow velocity was slow in the LAA in the HA model, and TAWSS was relatively small during one cardiac cycle (the upper-left in Fig. 8). On the other hand, the distributions of TAWSS in the AF_0 and AF_1 models hardly change in comparison with that in the HA model, but there are positions where values of the OSI and the RRT increase (Fig. 8). Especially on the LAA, the distributions of TAWSS are almost the same between the computational models, whereas the OSI and the RRT increase in the AF_1 model (Fig. 9). Consequently, the occurrence of AF promotes stagnation of blood flow by temporally changing flow directions of originally slow blood flow, resulting in a state in which thrombosis is easily created. This result agrees with the finding by CFD which clarified that disturbance of normal filling and emptying of blood by AF caused stagnation of blood flow in the LAA (Zhang and Gay, 2008). Interestingly, in the AF_1 model, positions with a large RRT are concentrated at the protrusions of the upper side of the LAA (Fig. 9). This implies that there is a local thrombus-prone site in the LAA where stagnation of blood flow is promoted when AF occurs.

A simplified deformation of the HA model mostly agreed with that of the actual LA observed by MRI (Fig. 2). Time variations of hemodynamics obtained by PC MRI and CFD were qualitatively the same, showing blood flow from PVs to the MV in diastole and a counterclockwise swirl in systole (Figs. 5 and 6). The E/A ratio of blood flow volume at the MV in the HA model (Fig. 4(a) and Table 1), which is considered to be an important criterion for diagnosis of heart function, was within the normal range of 1.5 ± 0.4 of the

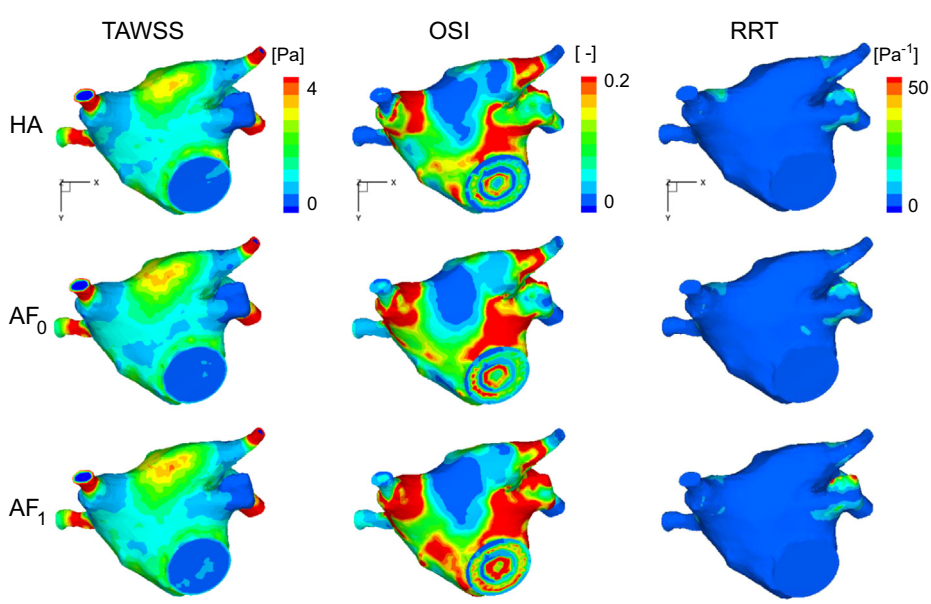

Fig. 8. Comparison of hemodynamic parameters (TAWSS, OSI, and RRT) on the LA between the healthy model (HA) and the diseased models (AF_0 and AF_1).

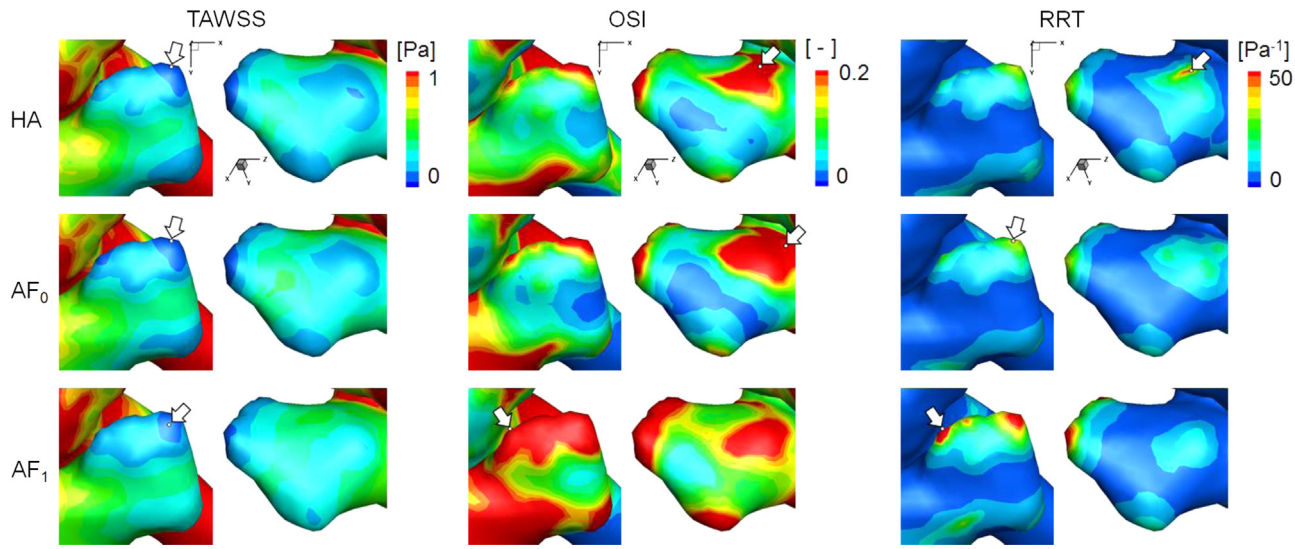

Fig. 9. Comparison of hemodynamic parameters (TAWSS, OSI, and RRT) on the LAA between the healthy model (HA) and the diseased models (AF_0 and AF_1). Arrows indicate the positions with minimum value of TAWSS ($TAWSS_{min}$), the maximum value of OSI (OSI_{max}), or the maximum value of RRT (RRT_{max}), respectively.

healthy LA (Appleton et al., 1988). Furthermore, deceleration time, which is used for evaluation of LV expansion, between the time of the maximum velocity at the E wave and the time of the minimum velocity just before the A wave was 236 ms in the normal range (Zile et al., 2001) (see a bottom line of a triangle of dashed lines in Fig. 4(a)). These imply that the application of deformation based on locations of the MV and the settings of boundary conditions are appropriate. In the diseased models (AF_0 and AF_1 models), hemodynamic changes in configuration and the number of vortices due to AF were confirmed as previously reported (Mouret et al., 2004; Zhang and Gay, 2008). In addition, clinically observed characteristic phenomena, such as the lack of the A wave in the blood flow volume at the MV (Fig. 4(a)) and the reduction of blood flow volume at PV during one cardiac cycle (Fig. 4(b) and Table 1), were reproduced (Nishimura et al., 1990).

However, several limitations remain to be considered. In this study, two conditions of opening and closing were assumed to instantly switch at the circular outlet of the MV, ignoring the real shape and movements of MV leaflets. The pressures at the LV in diastole and at the PV were assumed to be individually constant. Hence, a backflow at PVs owing to a sudden increase of LA pressure by the atrial kick was not reproduced (Pagel et al., 2003). Settings of pressure variation including how the MV opens/closes should be considered more accurately in future work. In the diseased models, AF was simulated by applying volume changes with reference to the locations of the MV annulus as a basal position. Since fluctuations of blood flow velocity at the MV observed in the AF_1 model (Fig. 4(a)) are hardly observed in clinical findings, it is necessary to consider more realistic conditions by applying fibrillations of different phases on the LA wall. Moreover, though morphology of the LA with AF does not show a significant change in the early stage of AF, it enlarges gradually. Therefore, the exact morphology of the diseased LA and connections of PVs to the LA (Dahl et al., 2012) should be considered in a patient-specific manner.

5. Conclusion

The effects of a lack of atrial kick and the occurrence of high-frequency fibrillation caused by AF on blood flow and hemodynamic stresses were quantitatively investigated by hemodynamic analyses using healthy and diseased LA models constructed based on MRI of a healthy volunteer heart. The results revealed that each characteristic phenomenon of AF influenced hemodynamics. Especially, the atrial wall movement by the high-frequency fibrillation has a large impact on stagnation of blood flow, increasing the RRT at the upper part of the LAA during AF. The result implies that there is a local thrombus-prone site in the LAA when AF occurs.

Conflict of interest statement

None of the authors have any financial and personal relationships with other people or organizations that could inappropriately influence (bias) their work.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.jbiomech.2014.12.025.

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